

Project Title: Dungeon of Darkness

CS20200 2026S Term Project

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Overview

Dungeon of Darkness is a 2D top-down hack-and-slash roguelike game. The player explores a dark dungeon, clears combat rooms, collects weapons, relics, and gold, and descends into deeper dungeon layers where enemies become stronger.

Requirements

1. Game World and Player

1. The game is played from a top-down view in a rectangular room.
2. Each room starts in a dark state.
3. The player can move, attack with the equipped weapon, and dash with a cooldown.
4. The player has HP. If HP reaches 0, the game ends.
5. The game displays the player's HP, gold, current room number, and current depth.
6. The game contains at least one playable character.
7. The player can equip exactly one weapon and hold multiple relics.

2. Rooms and Progression

1. The game consists of sequential rooms numbered from 1 upward.
2. Each combat room starts with a fixed or randomly selected group of enemies.
3. A room is cleared when all enemies in the room are defeated.
4. When a room is cleared, the room changes from dark to lit.
5. After clearing a room, the player chooses one reward from three displayed reward options.
6. After choosing a reward, the player enters the next room.
7. After every 8 cleared rooms, the player descends to the next depth.

3. Combat

1. Player attacks damage enemies inside the equipped weapon's hit area.
2. Each weapon has damage, attack cooldown, and hit area or range.
3. Enemies have HP, movement speed, and damage.
4. If an enemy or enemy projectile touches the player, the player takes damage.
5. Enemies are removed when their HP reaches 0.

6. The game contains at least 3 enemy type(s).

4. Weapons and Relics

1. The game contains at least 3 types of weapon.
2. Different weapons have visibly or mechanically different attack behavior.
3. The player can replace the equipped weapon when selecting or buying a weapon.
4. The game contains relics.
5. Each relic has a written effect.
6. Relics modify gameplay according to their written effects.
7. Selected or purchased relics become active immediately.

5. Gold and Shop

1. The player has a gold amount starting at 0.
2. Defeated enemies drop or award gold.
3. The player's current gold amount is displayed.
4. The game contains shop rooms or shop events that appear in room 4 of the depth.
5. A shop displays at least 3 purchasable options.
6. Shop options may include weapons, relics, HP recovery, or upgrades.
7. Each shop option displays its price.
8. The player may buy an option only if the player has enough gold.
9. Buying an option subtracts its price from the player's gold and applies the purchased effect.
10. The player may leave the shop and enter the next room.

6. Rewards

1. After each cleared combat room, the game displays three reward options.
2. The player selects exactly one reward.
3. A reward may be a weapon, relic, HP recovery, gold, or another clearly described upgrade.
4. The selected reward changes the player state according to its written description.
5. Unselected rewards are discarded.

7. Depth System

1. The depth starts at 0.
2. After 8 rooms, the depth increases by 1.
3. Enemy HP increases with depth.

4. Enemy damage increases with depth.
5. When a depth increases, the player must select from three debuff options.
6. The current depth is displayed to the player.

8. End Condition and Score

1. The game ends when the player HP reaches 0.
2. The final score is calculated from a fixed formula using the number of rooms cleared, the number of enemies slain, and remaining gold (formula will be displayed).
3. The game-over screen displays the final room count, depth reached, enemies slain, remaining gold, final equipped weapon, and number of relics held.
4. The player can restart the game or exit the program after the game is over.

9. Running Requirements

1. The game is implemented in F# using .NET 10.
2. The public GitHub repository includes the full source code.
3. The repository includes a README.md explaining how to build and run the game.
4. The game must be runnable by following only the README.md instructions.